Algebra: Applying the Zero Product Property

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DIRECTIONS

Solve each equation using the Zero Product Property: if a · b = 0, then a = 0 or b = 0. If the equation is not yet factored, factor it first, then set each factor equal to zero.

1. Solve:

$$(x-3)(x+5)=0$$

Answer: ______________________________

2. Solve:

$$x(x-7)=0$$

Answer: ______________________________

3. Solve:

$$(2x-1)(x+4)=0$$

Answer: ______________________________

4. Solve:

$$(3x+6)(x-2)=0$$

Answer: ______________________________

5. Solve:

$$x^2-9=0$$

Answer: ______________________________

6. Solve:

$$x^2-5x=0$$

Answer: ______________________________

7. Solve:

$$x^2+7x+10=0$$

Answer: ______________________________

8. Solve:

$$x^2-x-12=0$$

Answer: ______________________________

9. Solve:

$$2x^2+5x-3=0$$

Answer: ______________________________

10. Solve:

$$x^2+6x+9=0$$

Answer: ______________________________

Based on the Numberbender lesson "Applying the Zero Product Property (ZPP)" • youtu.be/BuOrHLgdKas

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Answer Key & Solutions

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TEACHER NOTES

Items 1–4: already factored — set each factor to zero. Items 5–6: factor out a common factor or use difference of squares. Items 7–9: factor a trinomial first. Item 10: perfect-square trinomial — double root.

1. Solve: $(x - 3)(x + 5) = 0$

Answer: $x = 3, -5$

$x - 3 = 0 \rightarrow x = 3$; $x + 5 = 0 \rightarrow x = -5$.

2. Solve: $x(x - 7) = 0$

Answer: $x = 0, 7$

$x = 0$ or $x - 7 = 0 \rightarrow x = 7$.

3. Solve: $(2x - 1)(x + 4) = 0$

Answer: $x = 1/2, -4$

$2x - 1 = 0 \rightarrow x = 1/2$; $x + 4 = 0 \rightarrow x = -4$.

4. Solve: $(3x + 6)(x - 2) = 0$

Answer: $x = -2, 2$

$3x + 6 = 0 \rightarrow x = -2$; $x - 2 = 0 \rightarrow x = 2$.

5. Solve: $x^2 - 9 = 0$

Answer: $x = \pm 3$

Difference of squares: $(x - 3)(x + 3) = 0 \rightarrow x = \pm 3$.

6. Solve: $x^2 - 5x = 0$

Answer: $x = 0, 5$

Factor x: $x(x - 5) = 0 \rightarrow x = 0$ or $x = 5$.

7. Solve: $x^2 + 7x + 10 = 0$

Answer: $x = -2, -5$

$(x + 2)(x + 5) = 0 \rightarrow x = -2$ or $x = -5$.

8. Solve: $x^2 - x - 12 = 0$

Answer: $x = 4, -3$

$(x - 4)(x + 3) = 0 \rightarrow x = 4$ or $x = -3$.

9. Solve: $2x^2 + 5x - 3 = 0$

Answer: $x = 1/2, -3$

$(2x - 1)(x + 3) = 0 \rightarrow x = 1/2$ or $x = -3$.

10. Solve: $x^2 + 6x + 9 = 0$

Answer: $x = -3$ (double root)

$(x + 3)^2 = 0 \rightarrow x = -3$.

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